AI101 Mini-Project

PREDICTING CELLULAR NETWORK SIGNAL STRENGTH FROM ENVIRONMENTAL AND NETWORK PARAMETERS

Using Linear Regression

Team no. 59

Members:

Sumeksha -2026EEB1416	PPT Lead
Palak Tripathi-2026EEB1388	Video Lead
Sumit2026MCB1574	Documentation Lead
Aadarsh Aawalasiya- 2026AIB1004	Execution Lead

Repo link:https://github.com/2026mcb1574-pixel/ai-project-linear-regression.git

What and why

- Predicting the signal strength (dBm) of a mobile connection from network parameters (SNR, attenuation, distance to tower(km)) and the surrounding environment (urban, suburban, open, home)
- Why it matters: weak signal means dropped calls and slow data. Knowing what drives signal strength helps operators place towers and improve coverage.

463
rows

11
columns

Target: Signal Strength (dBm)
Features: SNR, Attentuation, Distance to Tower(km), Environment (open/ urban/ suburban/home**)**

Dataset source

Kaggle:
“cellular-network-performance-data” by Suraj (suraj720)
https://www.kaggle.com/datasets/suraj520/cellular-network-performance-data

Approach

1 → 2 → 3 → 4 → 5 → 6

1 Data

Kaggle dataset(train.csv), 463 rows,11 columns

2 Preprocessing

- Removed duplicate rows
- Converted target to numeric (invalid values → NaN) and dropped rows with missing target
- Filled missing values: median for numeric columns, most frequent for Environment
- One-hot encoded Environment (unseen categories ignored)

3 Visualization

Explore the data with graphs:

Histogram, Scatter Plots(distance, SNR), Box Plot by Environment, Signal Quality Class Chart

4 Training

- 80/20 split (test_size=0.2): 370 training and 93 testing examples
- Pipeline= Preprocessing + LinearRegression, fitted on training data only(no data leakage)

5 Testing and Error Calculation

predicted on the 93 unseen test examples and evaluated with R^2 score, MAE,MSE and RMSE

6 Output

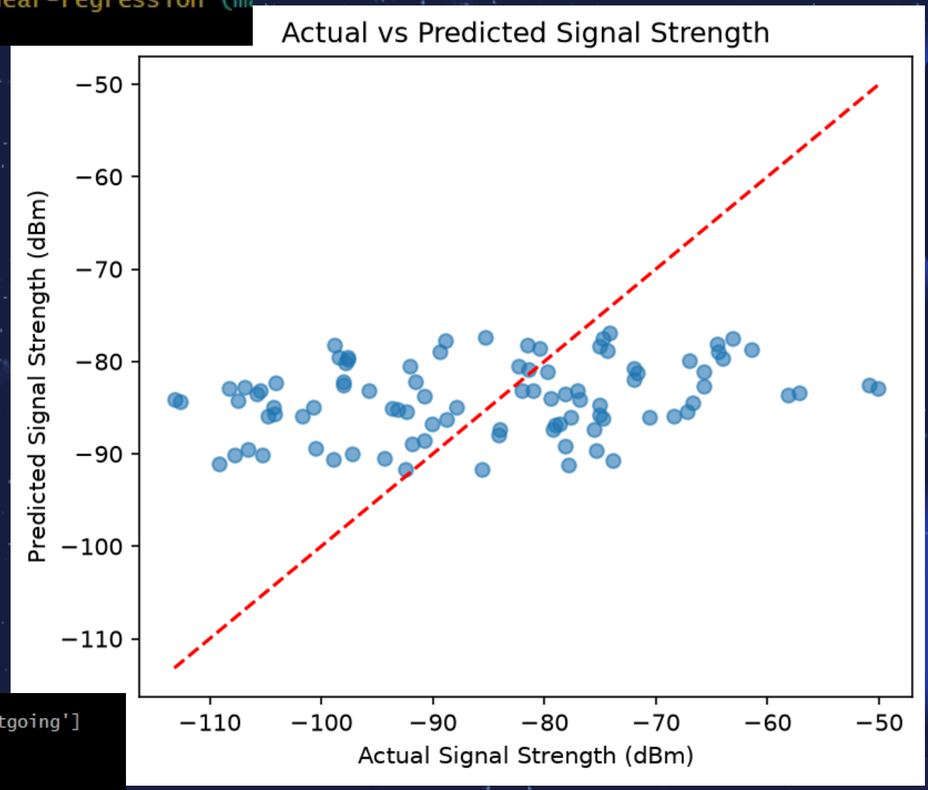
predicted signal strength (dBm) for each testing example, compared with actual values in an actual vs predicted plot

Libraries used: Pandas, Numpy , Scikit-learn (LinearRegression, OneHotEncoder, ColumnTransformer,Pipeline,SimpleImputer,train_test_split,metrices), Matplotlib

RESULTS

```
Jyoti@LAPTOP-S8R95H58 UCRT64 /d/Users/Jyoti/Downloads/AI mini project Execution
$ git clone https://github.com/2026mcb1574-pixel/ai-project-linear-regression.git
Cloning into 'ai-project-linear-regression'...
remote: Enumerating objects: 124, done.
remote: Counting objects: 100% (124/124), done.
remote: Compressing objects: 100% (106/106), done.
remote: Total 124 (delta 46), reused 29 (delta 11), pack-reused 0 (from 0)
Receiving objects: 100% (124/124), 490.82 KiB | 3.48 MiB/s, done.
Resolving deltas: 100% (46/46), done.
cdcd
Jyoti@LAPTOP-S8R95H58 UCRT64 /d/Users/Jyoti/Downloads/AI mini project Execution
$ cd ai-project-linear-regression

Jyoti@LAPTOP-S8R95H58 UCRT64 /d/Users/Jyoti/Downloads/AI mini project Execution/ai-project-linear-regression (ma
$ pip install -r requirements.txt
```

```
Requirement already satisfied: pure-eval in C:\Users\Jyoti\AppData\Roaming\Python\Python314\site-packages (from stack_data>

Jyoti@LAPTOP-S8R95H58 UCRT64 /d/Users/Jyoti/Downloads/AI mini project Execution/ai-project-linear-regression (main)
$ python linear-regression.py
Dimensions of dataset: (463, 11)

Column names:
['Timestamp', 'Signal Strength (dBm)', 'SNR', 'Call Duration (s)', 'Environment', 'Attenuation', 'Distance to Tower (km)',

   Metric       Value
      MAE   12.297309
      MSE  210.444819
     RMSE   14.506716
 R2 Score    0.067065
 Actual Signal Strength  Predicted Signal Strength  Error (Actual - Predicted)
             -82.060041                -83.162773                    1.102732
             -75.037626                -85.846010                   10.808384
             -88.904164                -77.752349                  -11.151815
             -90.745061                -83.714138                   -7.030923
             -66.710577                -84.485327                   17.774750
             -87.866356                -84.924503                   -2.941854
             -91.892598                -88.892639                   -2.999960
             -79.249967                -87.407585                    8.157618
             -66.972822                -79.912457                   12.939635
            -104.138485                -85.721608                  -18.416878

oti@LAPTOP-S8R95H58 UCRT64 /d/Users/Jyoti/Downloads/AI mini project Execution/ai-project-linear-regression (main)
python data_visualization.py
lass 'pandas.DataFrame'>
ngeIndex: 463 entries, 0 to 462
ta columns (total 11 columns):
Target Column: Signal Strength (dBm)
Feature Columns: ['SNR', 'Call Duration (s)', 'Environment', 'Attenuation', 'Distance to Tower (km)', 'Call Type', 'Incoming/Outgoing']
   Signal Strength (dBm)        SNR  ...  Call Type Incoming/Outgoing
0            -84.119763  25.940082  ...       data          incoming
1            -87.806131  15.937354  ...      voice          incoming
2           -116.575176  14.709384  ...      voice          incoming
3            -82.961183  21.727165  ...       data          outgoing
4            -85.012574  26.058158  ...      voice          outgoing

[5 rows x 8 columns]
Input shape (X): (463, 7)
Target shape (y): (463,)
Numeric features: ['SNR', 'Call Duration (s)', 'Attenuation', 'Distance to Tower (km)']
Categorical features: ['Environment', 'Call Type', 'Incoming/Outgoing']
Rows available for modeling: 463

Jyoti@LAPTOP-S8R95H58 UCRT64 /d/Users/Jyoti/Downloads/AI mini project Execution/ai-project-linear-regression (main)
$
```



210.445
MSE

12.297
MAE(dBM)

14.507
RMSE (dbM)

0.067
R^2 Score

*All values (R^2 score, MAE,MSE,RMSE) are rounded off upto 3 decimal places

Learnings

What worked

- Clean pipeline: preprocessing → training→ testing → error calculation
- Imputation and One-hot encoding handled missing and categorical data
- Branches and pull requests kept our work organized
- Tasks were split clearly across all four members
- Graphs showed the patterns in the data (for example the effect of Environment) before modelling

What didn't work

- A hardcoded local file path(C:/Users..) broke the notebook on other machines twice and was fixed with relative paths.
- Some early commits went straight to main branch, so we moved to Pull Requests
- Low R^2 (0.067) and high RMSE (14.5 dBm) : distance and SNR showed almost no linear relationship with signal strength

Who did what

- Sumeksha- Preprocessing and model training
- Aadarsh Aawalasiya- Data Visualization (graphs)
- Palak Tripathi- Model Testing
- Sumit- Error calculation (MAE, MSE, RMSE, R^2 score)

AI-use disclosure

We used Claude, Chatgpt (AI assistants) for help with the Git/GitHub workflow, debugging errors such as the file paths.